

SHARE Seattle 2010

2460: DFSMS Basics: The ABC's of DFSMSrmm Part 1 of 2

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A functional introduction to key RMM functions; how they work and what they deliver for your benefit. Learn how to define and manage the RMM resources and the options available to you.

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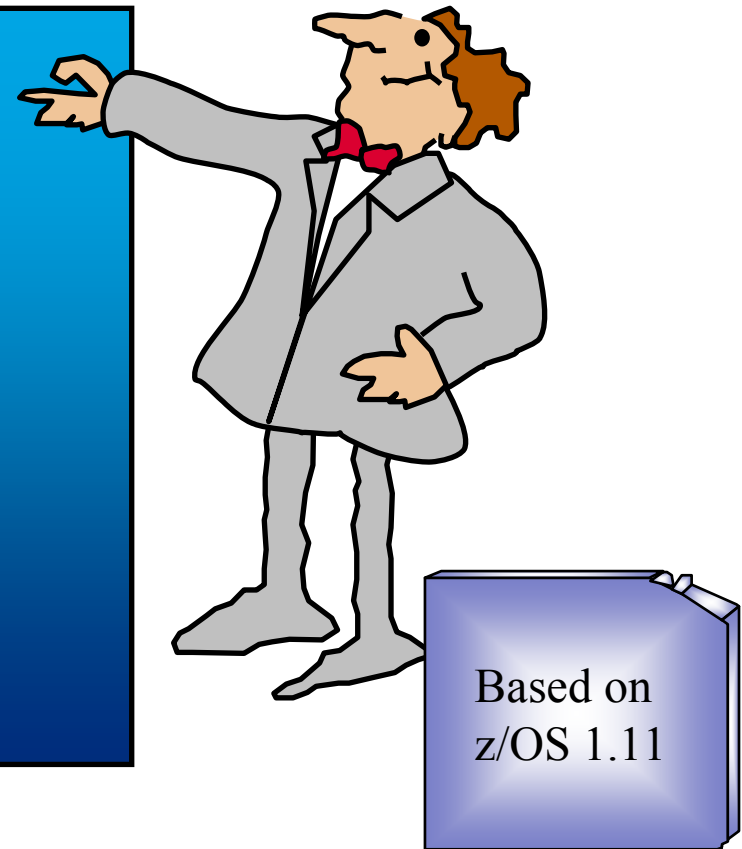
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Agenda

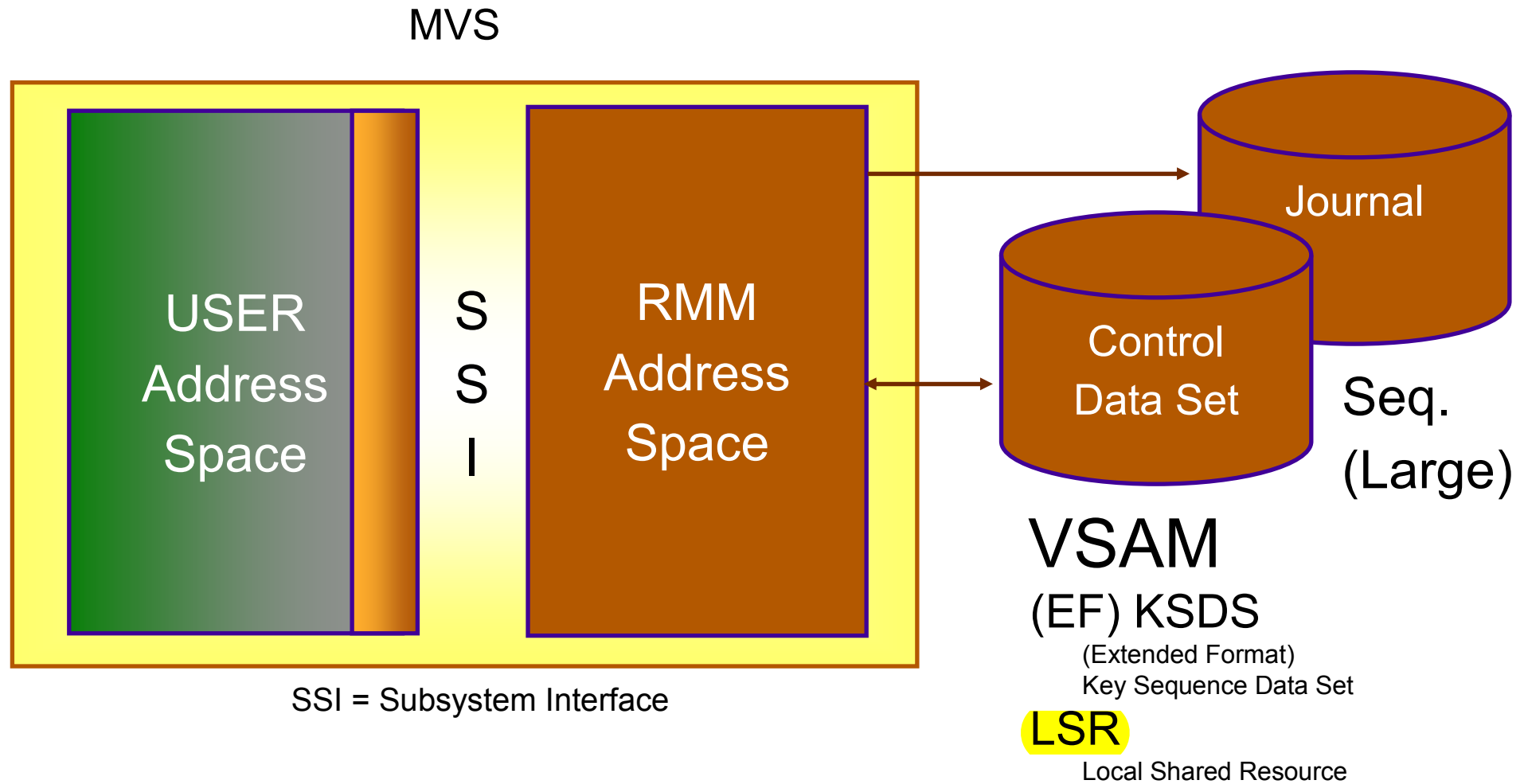
- Introduction to RMM
- Resources managed by RMM
- Software Interfaces
- User/Program Interfaces
- Inventory Management
- Tools



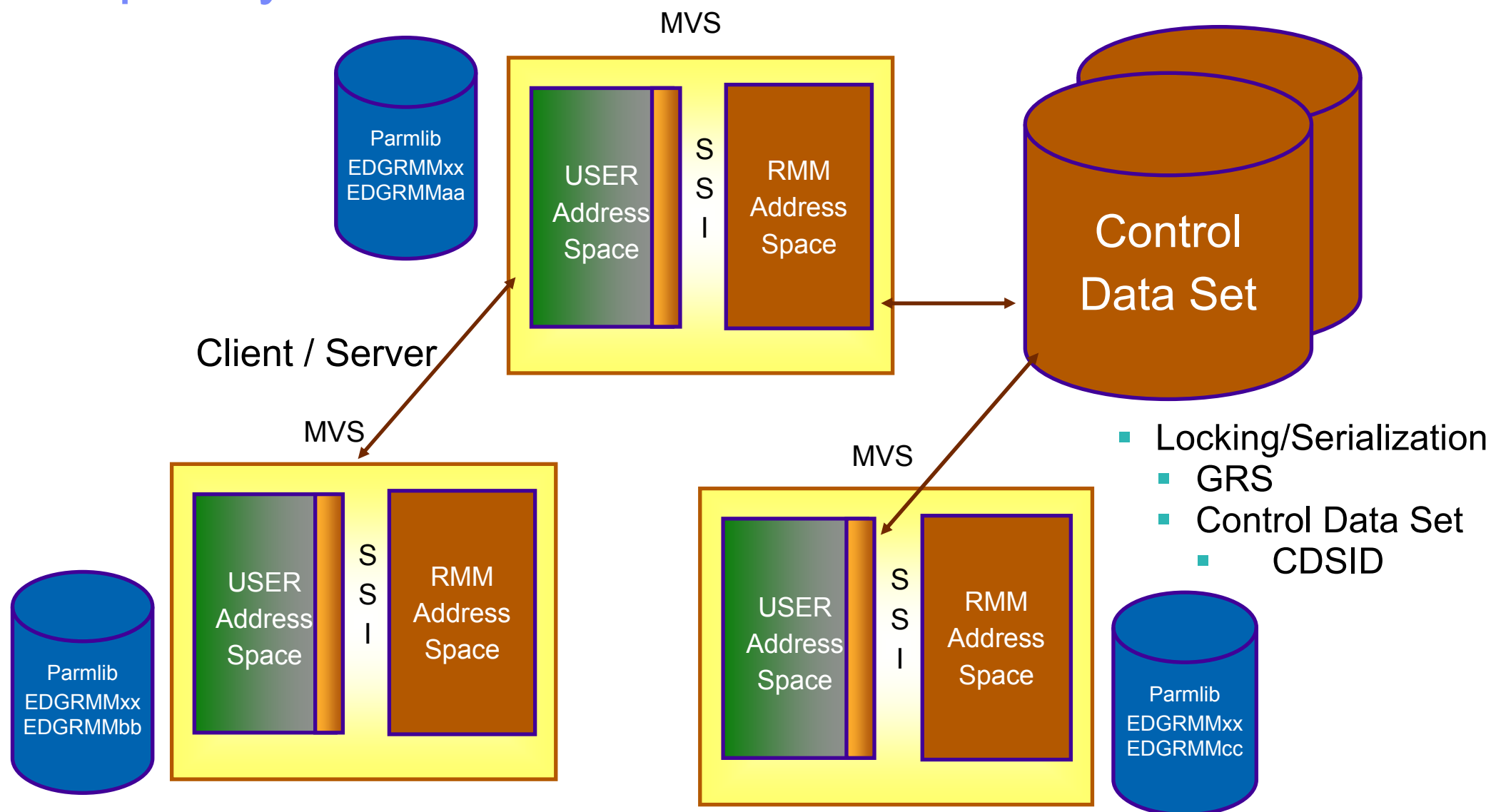
our product ...DFSMSrmm

- Removable Media Manager
 - IBM's tape management system for z/OS
 - A complete, centralized and automated system
 - Supports all tape devices and tape libraries
- Protecting tape data sets from accidental destruction
- Manage volume retention and movement
- Support and enhance system-managed tapes
- An Element of z/OS

Structure...

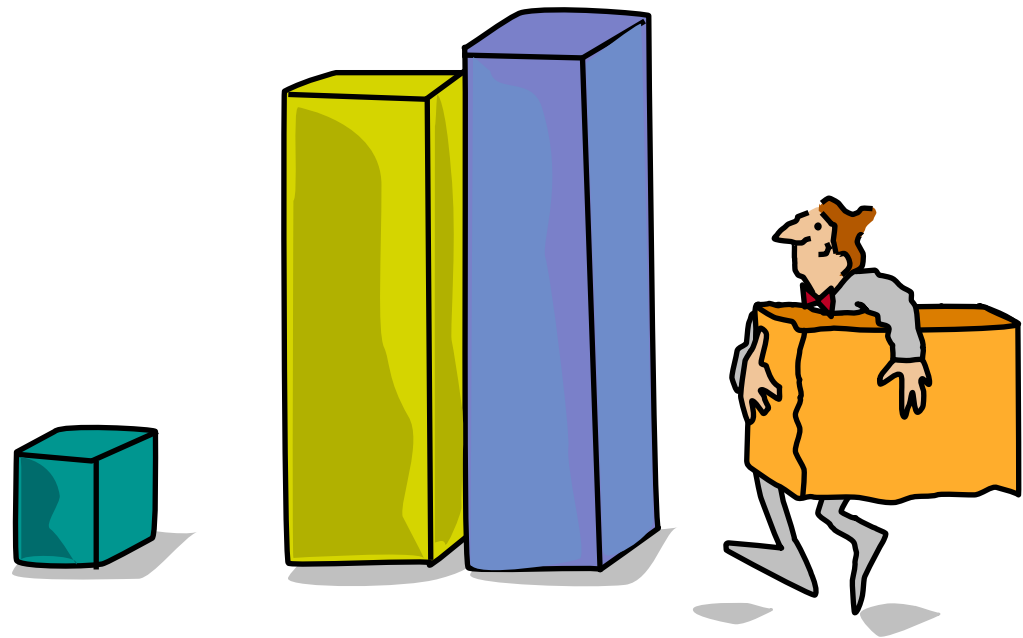


Multiple Systems ...

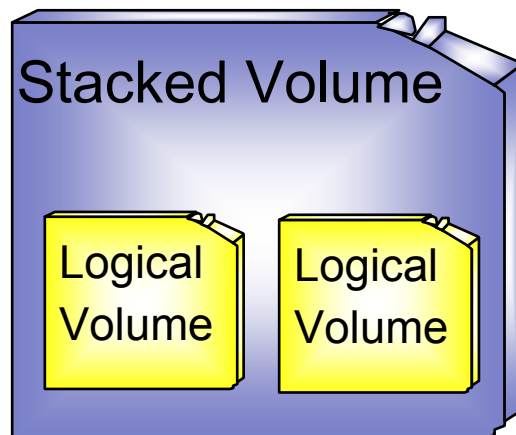
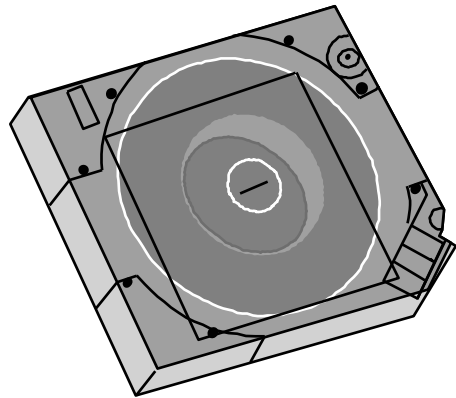


RMM managed resources

- Volume
- Data Set
- Owner
- Location
- Rack / Bin
- Vital Record Specification
- Software Product



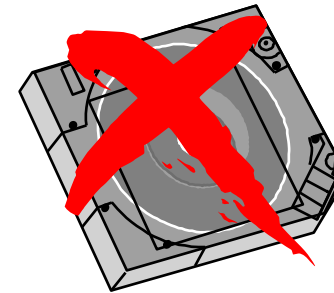
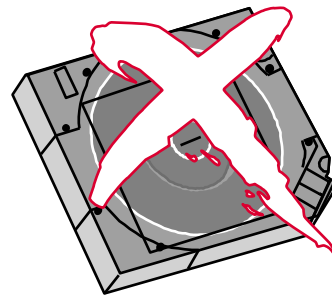
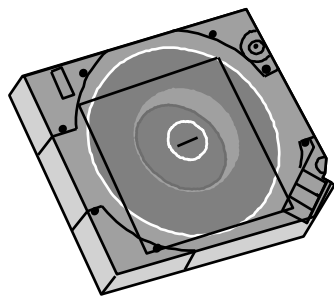
Volume ... Types



- Physical Volume
 - WORM WWUCID
- Logical Volume
 - Used in a Virtual Tape Server
 - Logical Volume / Virtual Volume
- Stacked Volume
 - Container for exported Logical Volumes (VTS)
 - Container for Copy Export
 - Box for Physical Volumes

Volume ...

- Volume Status
 - Master
 - Scratch
 - User
- Availability
 - Pending Release
 - Vital Record retained
 - On Loan
 - Open
- Expiration Date
- Retention Date



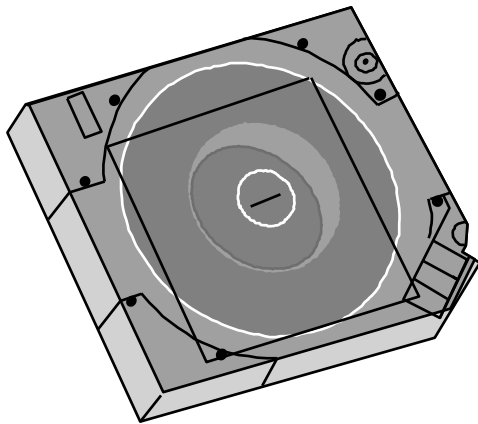
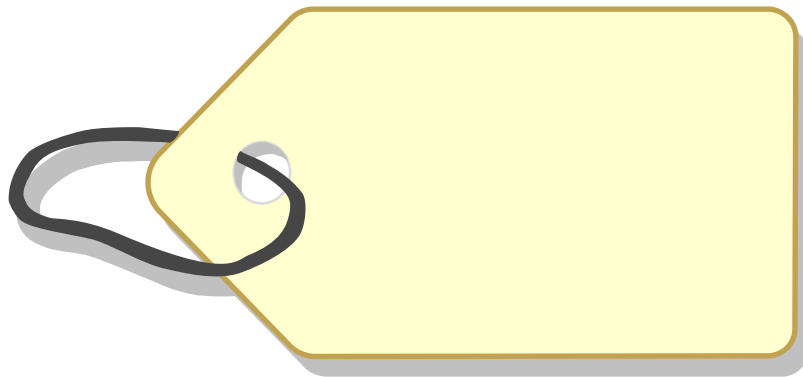
Volume ...

Validation

- Non specific mount request
 - Only scratch volumes can be mounted
- Specific mount request
 - The correct, non-scratch volume must be mounted
- PRTITION and OPENRULE commands in parmlib
 - Subset volumes and processing options
- When a data set is opened and closed DFSMSrmm automatically updates the volume and data set record



Volume ...



- Label
 - Standard Label
 - ANSI Label
 - No-Label

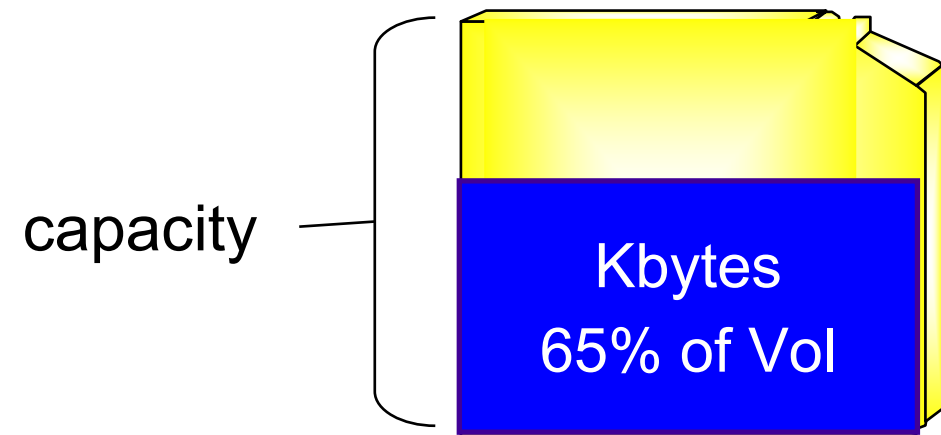
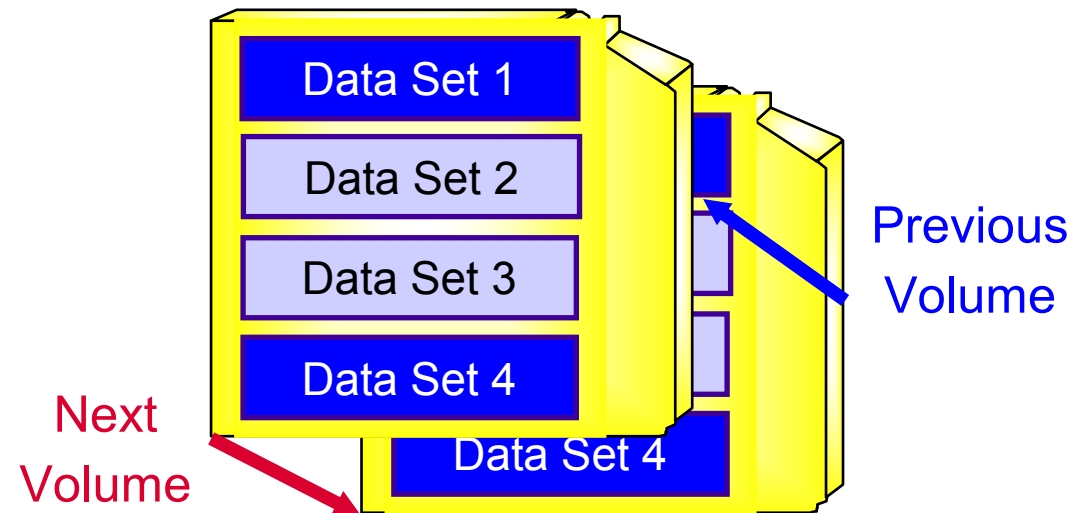
**duplicate
VOL1
supported**

- Media Attributes
 - Type
 - Capacity
 - Recording format
 - Compaction

**MEDINF
for non-
IBM media**

Volume ...

- Data Sets on Volume
 - First DSN
 - Number of data sets
- Multi Volume Set
 - Next Volume
 - Previous Volume
 - Volume Sequence
- Volume usage
 - Application written
 - Hardware recorded

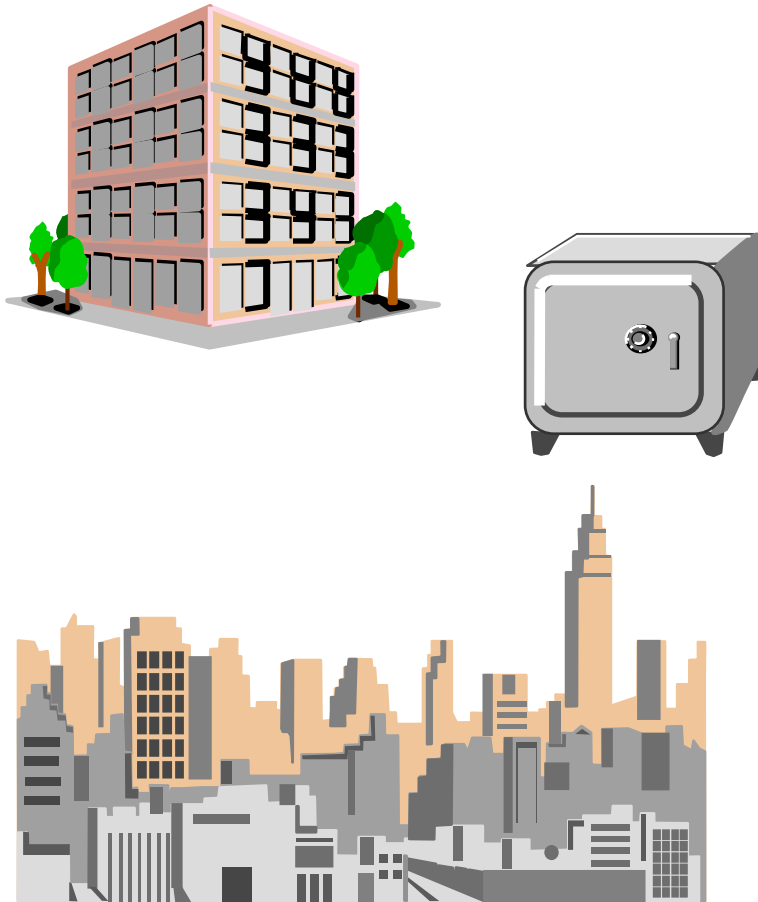


Volume ... Security



- Security class
- Classification
- Owner
 - Owner Access
- Volume Access List

Volume ... Location



- Current Location
- Home Location
- Required Location
- Destination Location
- Old Location
- Loan Location
- Rack / Bin
- In Transit Status

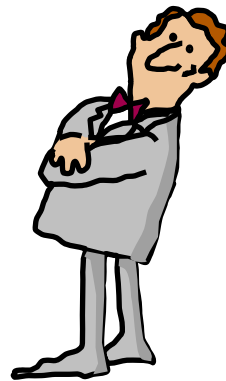
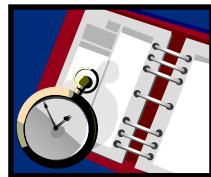
Volume ... Tracking

- Last Change

- Date/Time

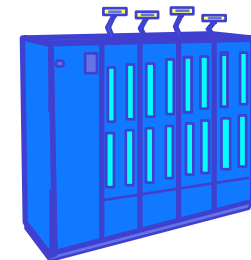
- Userid

- Systemid

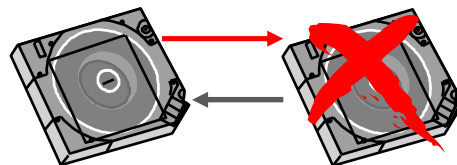


- Last Read Date

- Last Write Date



- Status Change Date/Time



- Drive last used

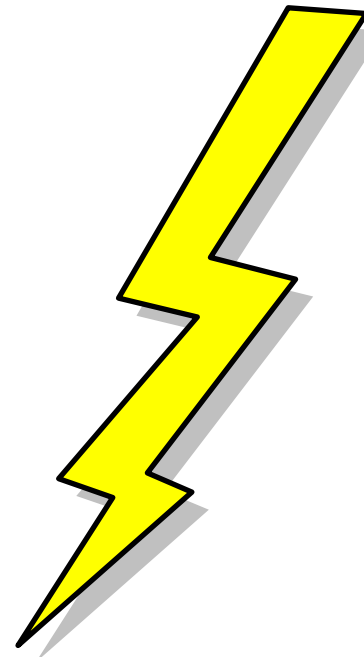
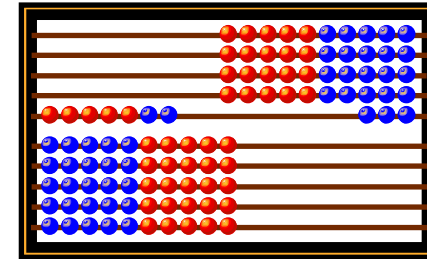
- Last Move Date

- Write Mount Count

Volume ... Tracking

- Volume use count
- Write Mount count

- Errors
 - Temporary
 - Read
 - Write
 - Permanent
 - Read
 - Write

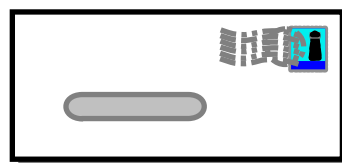


Volume ... Release Actions

- Return to Scratch



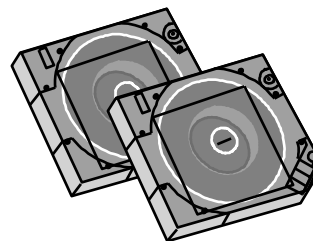
- Notify Owner



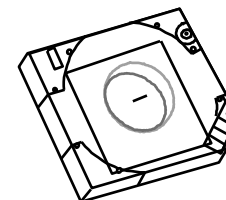
- Return to Owner



- Replace

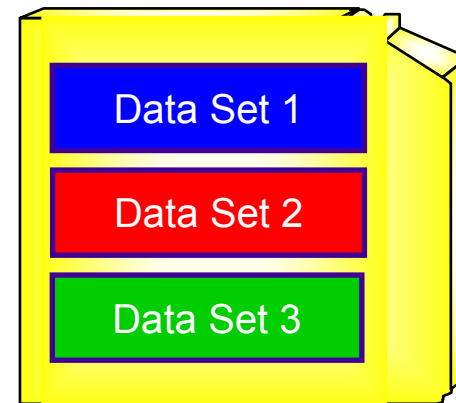


- Init / Erase

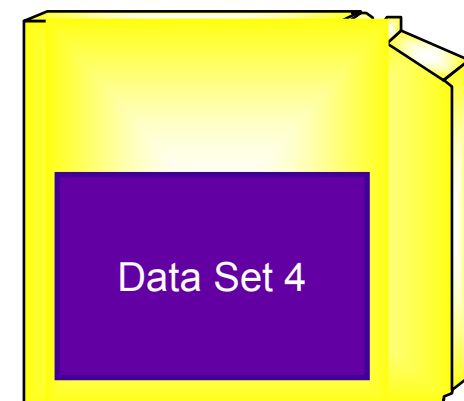
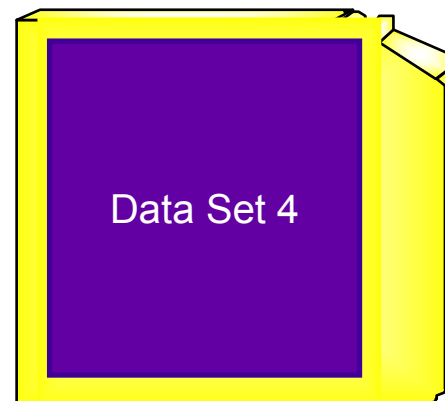


Data Set ...

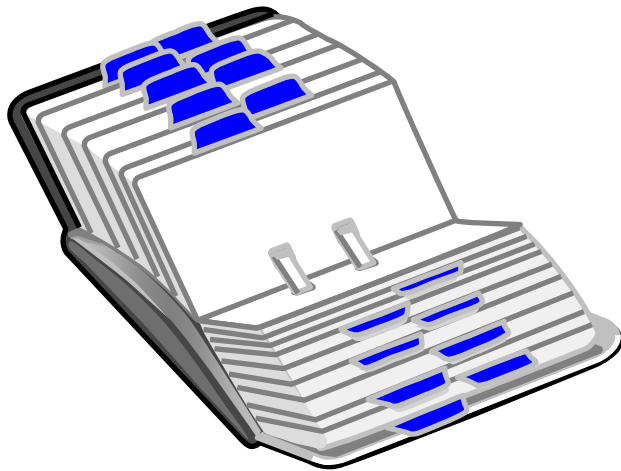
- Multiple Data Sets on one Volume



- One Data Set on multiple Volumes



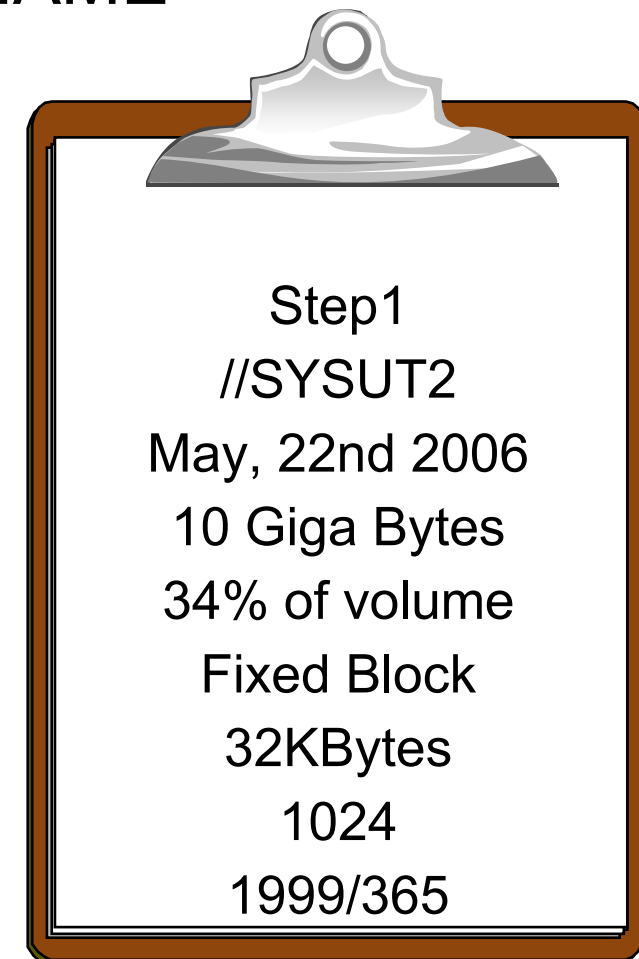
Data Set ... Control Info



- Key
 - Data Set Name (44 Character)
 - Volume Serial
 - Physical File Sequence Number
- Job Name
- Management Class
- Management Value

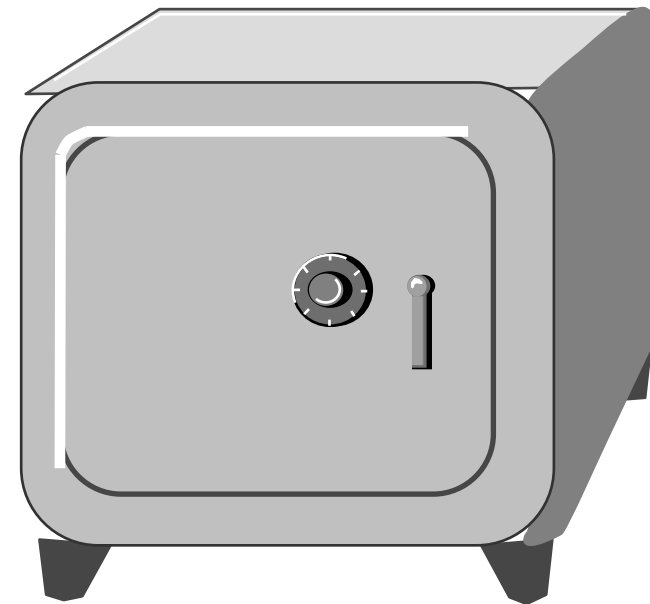
Data Set ... Attributes

- Step Name / Program Name / DD NAME
- Create Date / Time
- Date last read / written
- Data Set Size
- Record Format
- Block Size
- Logical Record Length
- EXPDT and Original EXPDT
- ABEND, DELETED



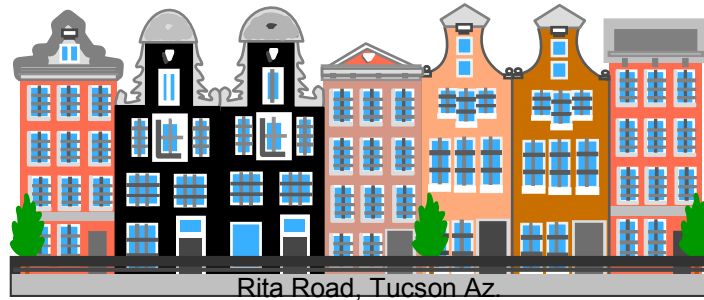
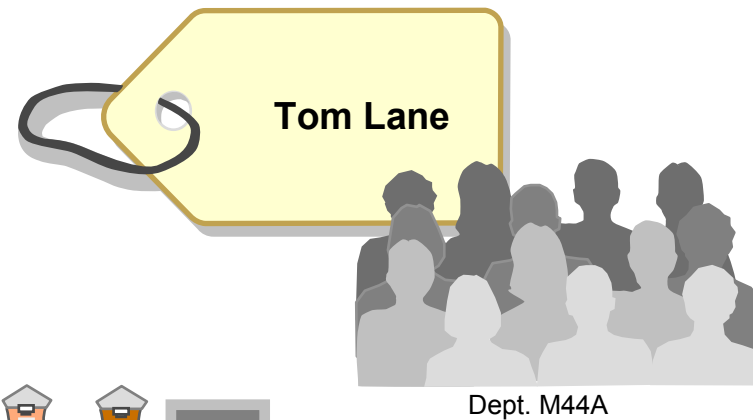
Data Set ... Vital Record

- Vital Record Status
- Matching Vital Record Specification
- Current VRS Sub-Chain
 - name
 - date
- Retention Date

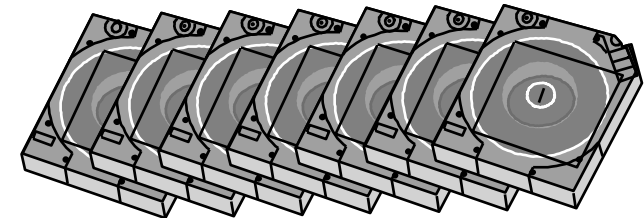


Owner ...

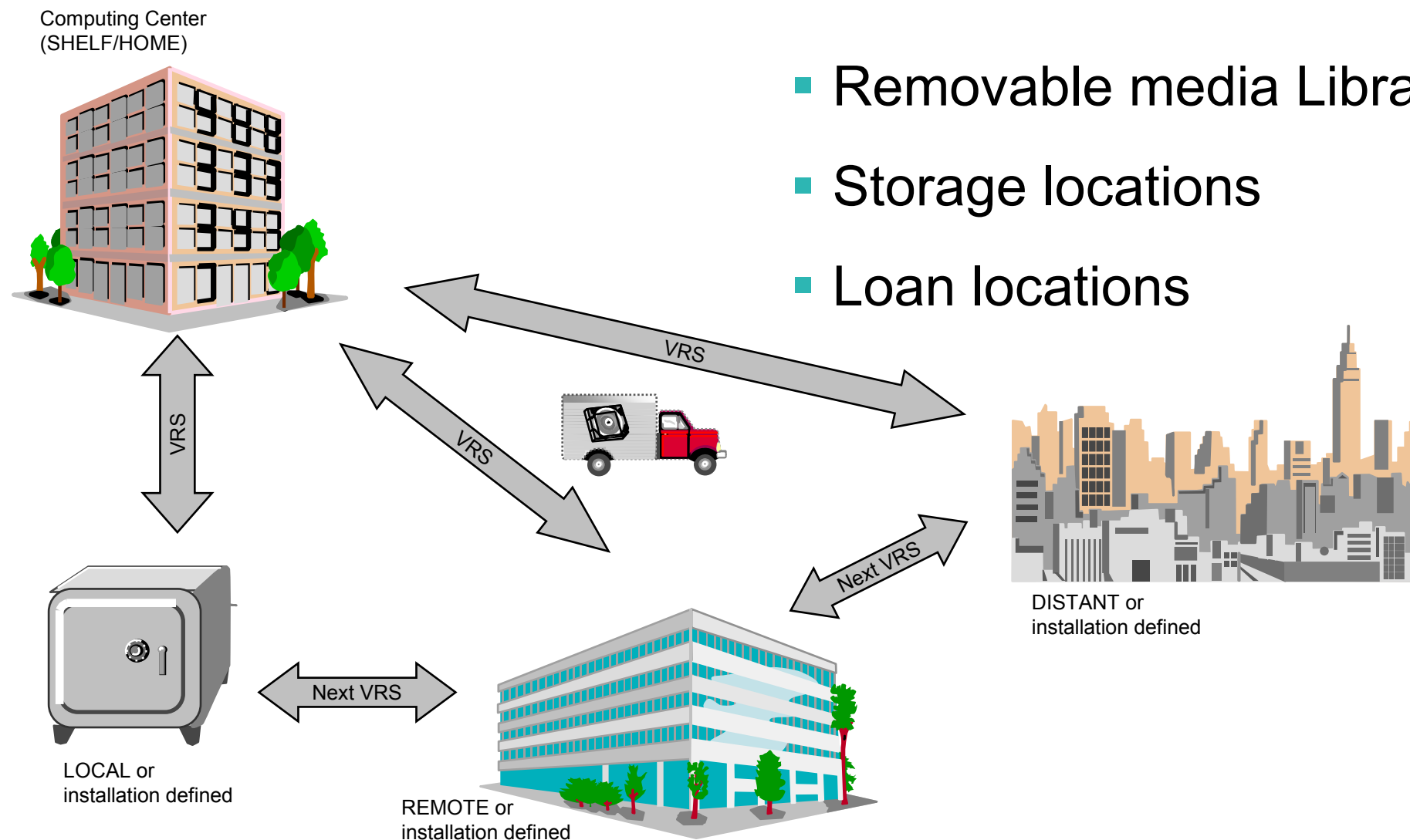
- Name, Department, etc.
- Address
- E-mail
 - user@hostname or Node.Userid



- List of owned Volumes



Location...



Location... shelf Space

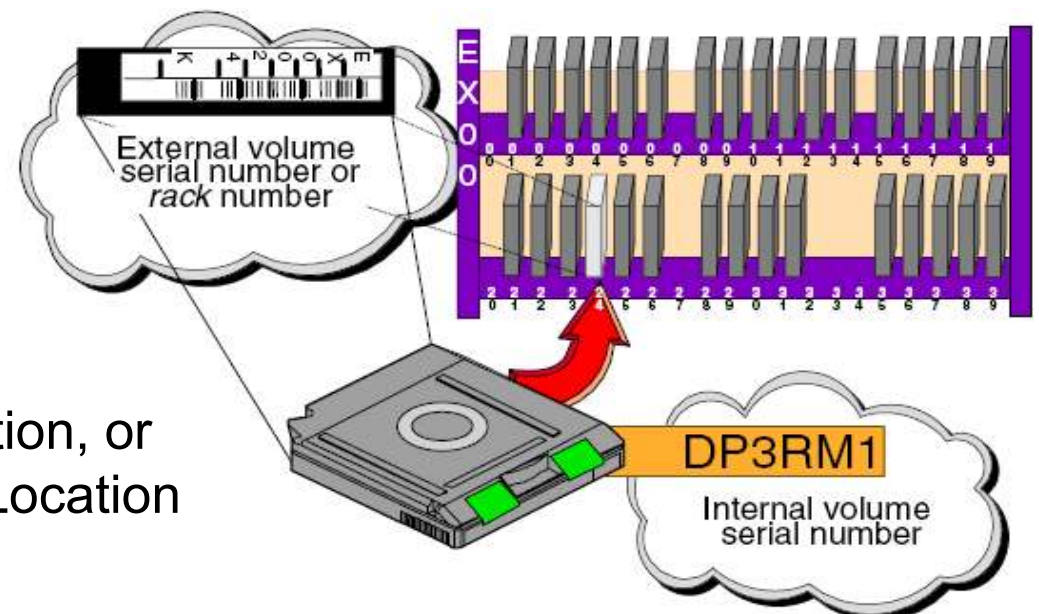
■ Rack Number

- Optional for any Volume
- Assigned slot for volume storage
- Can be freed and re-assigned via commands
- Grouped into pools using VLPOOL PREFIX(pref*)

• Selected using
MEDIANAME

■ Bin Numbers

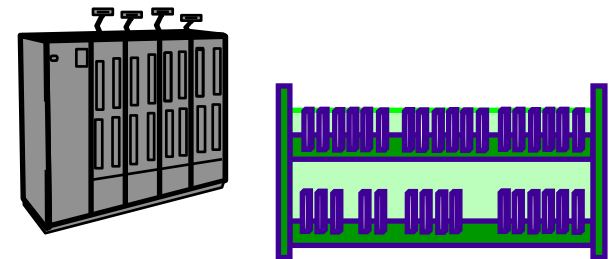
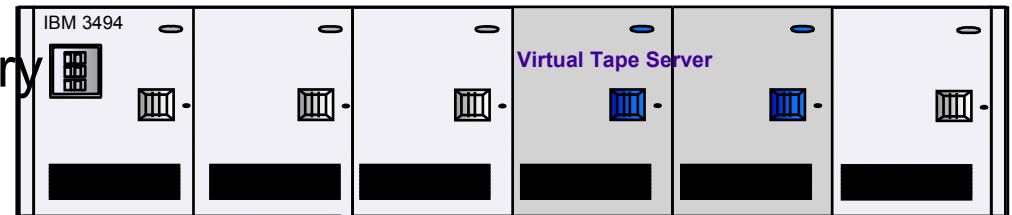
- Optional by Storage Location
- Assigned at start of a move
- Reusable either:
 - on CMOVE out of Storage Location, or
 - at Start of Move out of Storage Location



Location ...

System –Managed Tape Library

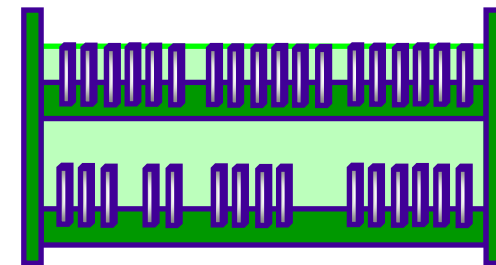
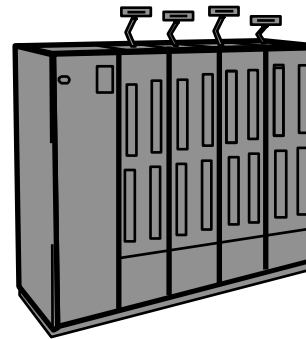
- Automated
 - IBM TS7700 Virtualization Engine
 - IBM TS3500
 - IBM 3494 Enterprise Tape Library
 - IBM Virtual Tape Server ...
- Manual
 - A set of tape drives defined as a Library
- Location Name: library_name



Location...

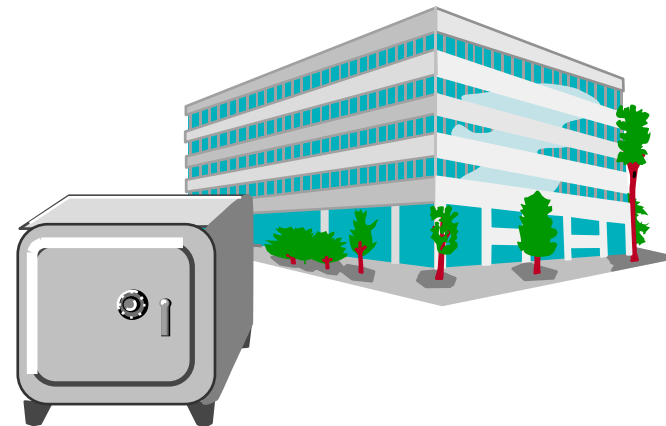
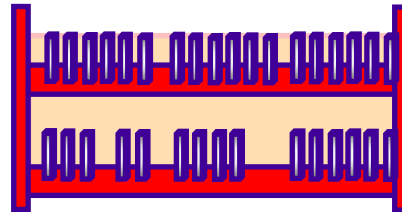
Non System-Managed Tape Library

- Traditional Tape Library
- Other Robots
- Location Name:
 - SHELF
 - Any Store Name
- Rack management supported



Location... Storage Location

- Management
 - No Bins
 - Bins
 - Media name
- Installation defined Storage Location
 - Unlimited number, any name
- Built-in Storage Locations
 - Location Name: LOCAL
 - Location Name: REMOTE
 - Location Name: DISTANT

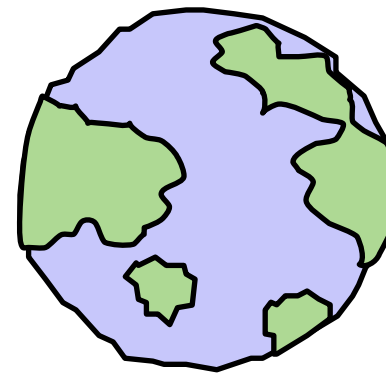


Vital Record Specification

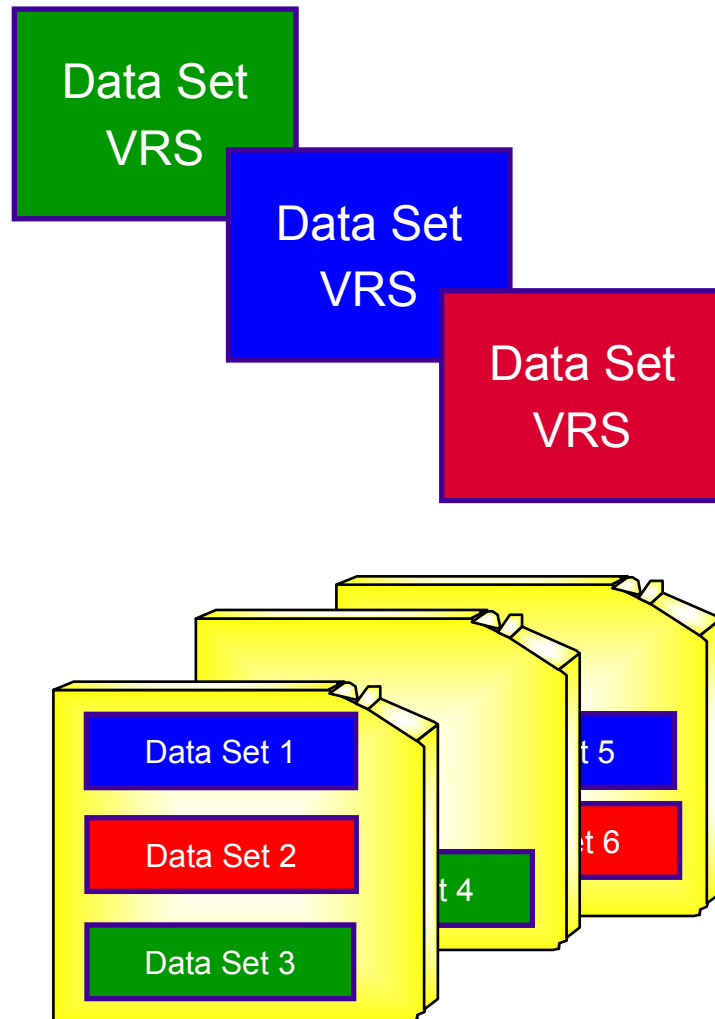
- VRS's are Policies to ...
 - control Retention of Data Sets and Volumes
 - determine Volume's required Location
 - Set Release Options



- **VRSEL Processing**
 - Applies Policies

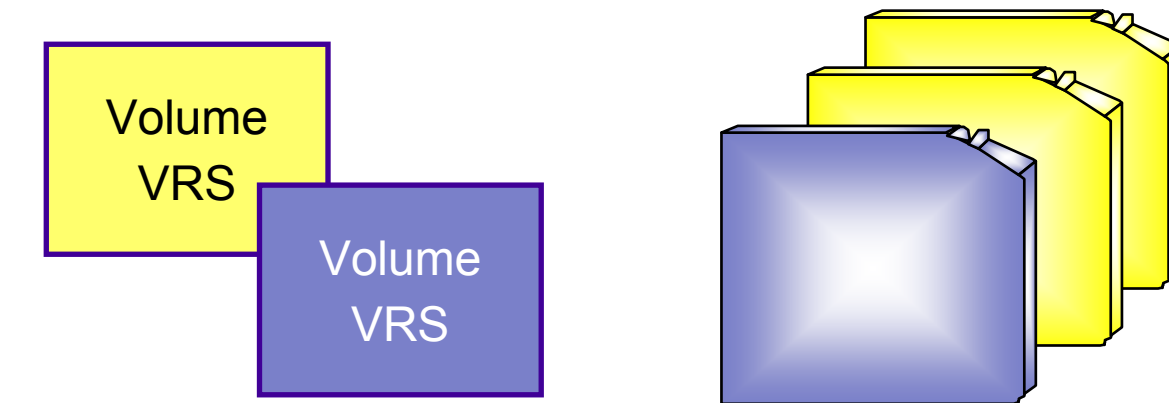


Vital Record Specification

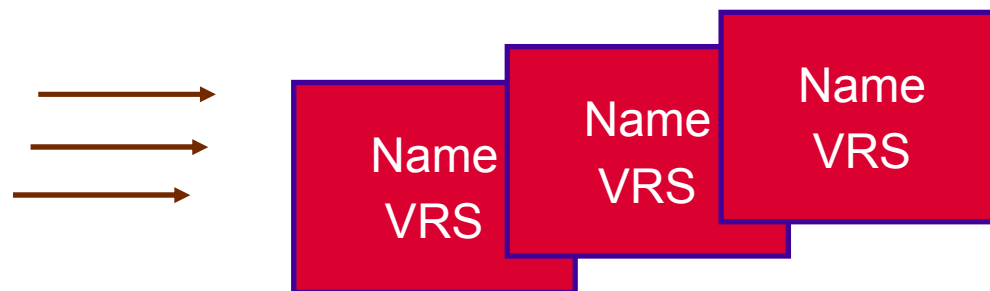


- Data Set VRS
 - Select Data Sets by:
 - OPEN flag
 - DELETED flag
 - ABEND flag
 - Management Class
 - Management Value
 - Data Set Name
 - Jobname
 - Generation Data Group
 - Generic Masks supported
 - Plus - Pseudo GDG

Vital Record Specification



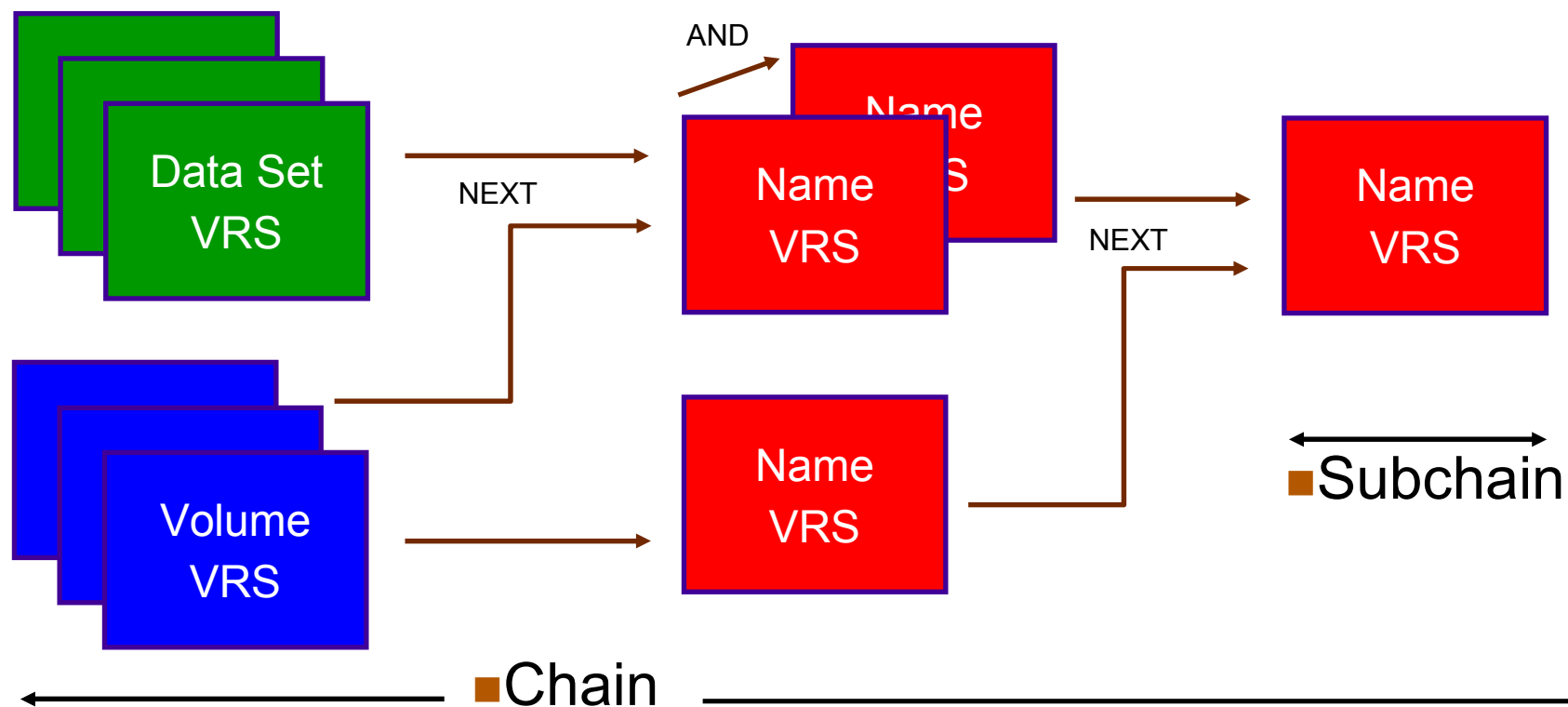
- Volume VRS
 - Specific
 - Generic



- Name VRS's
 - Location only
 - Retention & Location

Vital Record Specification

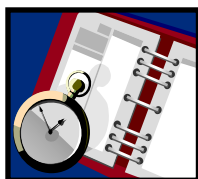
- Data Set VRS
- Volume VRS
- Name VRS



Vital Record Specification

Retention Types

- Specific Date



- Cycles

- By Days Cycle



- While Cataloged

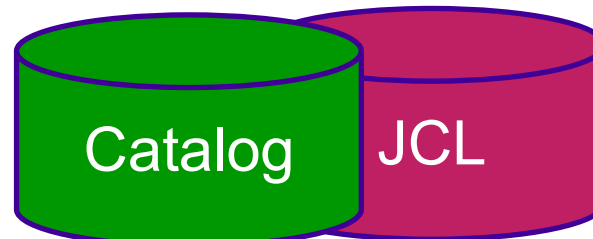
- Until Expired

- Days since creation

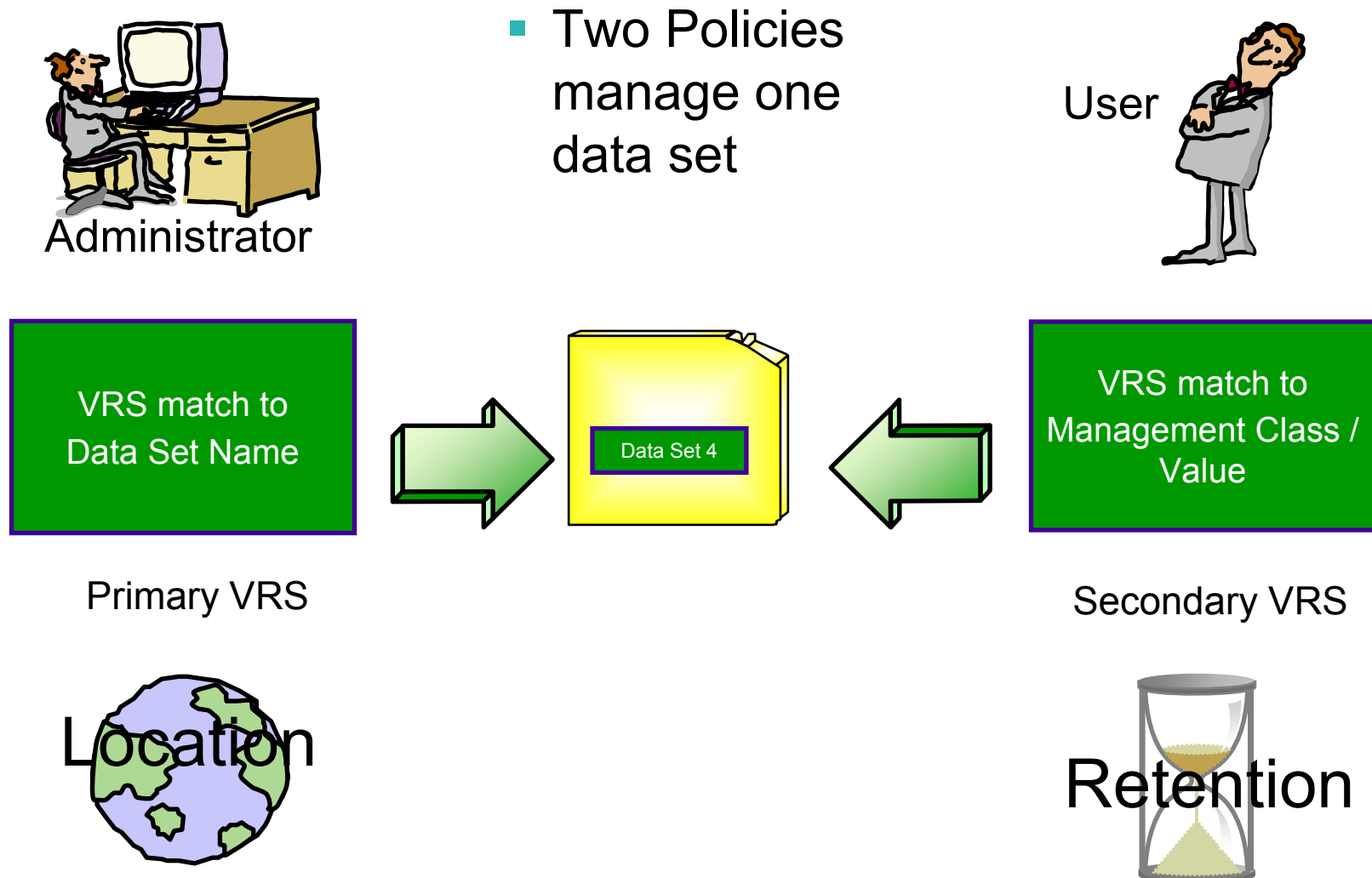
- Days since last reference

- Extra days

- Forever



Vital Record Specification



Audit Controls for VRS and Release Processing

- **Avoiding Loss of Data**

- Pre-empt system errors, user errors
- Define expected retention as an overall control
- Exploit automated way, to detect when too many volumes are released or expired

- **New parmlib options are added**

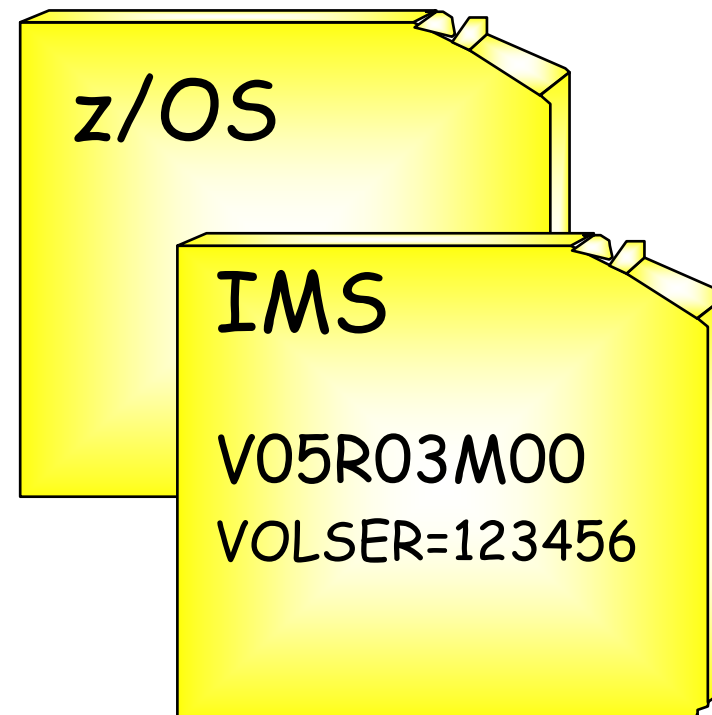
- Use VRSDROP to control volumes dropped from VRS retention
- Use VRSRETAIN to control newly assigned volumes that must be retained by VRS
- Use EXPDTDROP to control volumes dropped from EXPDT retention

- **New Analysis is available**

- EDGJACTP sample helps report/analyze exceptions

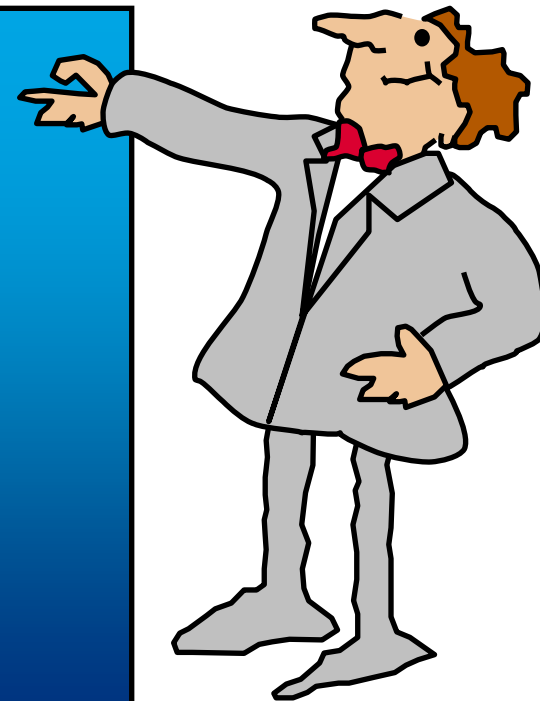
RMM managed resources

- Volume
- Data Set
- Owner
- Location
- Rack / Bin
- Vital Record
- Specification
- **Software Product**



Agenda for Part II

- Introduction to RMM
- Resources managed by RMM
- Software Interfaces
- User/Program Interfaces
- Inventory Management
- Tools



Summary

Protect
Data on Tape

Enforce
Security

Control
Retention

Support
Automation

Manage
Vaulting



Thanks for your interest
in
DFSMSrmm ...

See you for Part II

Removable Media Manager